

Gaurav Kumar

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EDUCATION

Guru Ghasidas Vishwavidyalaya, Bilaspur, Chhattisgarh November 2022 – May 2026
B.Tech. in Electronics and Communication Engineering CGPA: 8.85/10.0

Sri Chaitanya Techno School, Visakhapatnam May 2018 – March 2020
Intermediate (PCM) Percentage: 85/100

Relevant Coursework: Network, Signals and Systems, Electronic Devices and Circuits, Digital Electronics, Communication, Analog Electronics, CMOS VLSI Design, Linear Integrated Circuits

Trainings & Workshops: RTL Design Workshop (KT Semicon), RTL Design and Verification (The Silicon Sandbox), VLSI for Beginners (NIELIT Calicut)

SKILLS

Programming & Hardware Languages: Verilog HDL, Python, MATLAB

VLSI & Digital Design: RTL Design, CMOS Design, FSM Design, Timing Analysis, Functional Verification

Engineering Tools & Simulation: Xilinx Vivado, Cadence Virtuoso, LTspice, Proteus, Simulink, GitHub, VS Code, Jupyter Notebook

Web & Design Technologies: LaTeX, Overleaf, WordPress, Netlify, Adobe Illustrator, Canva

PROJECTS

SRAM & DRAM Design and Analysis | *Verilog HDL, Cadence Virtuoso, Xilinx Vivado* April 2026

- Implemented and conducted analysis of **SRAM** and **DRAM** memory architecture and cell view using **Verilog HDL** in **Xilinx Vivado** and **CMOS Design** using **45nm technology** in **Cadence Virtuoso**.
- Developed **SRAM** and **DRAM** architecture, **timing diagrams**, and conducted **read/write operation analysis** for understanding memory behavior.
- Designed **6T SRAM Cell** and **1T1C DRAM Cell** in **Cadence Virtuoso** and analysed **DC** and **Transient behaviour** for read/write operations at **1.8V** supply.

Traffic Light Controller | *Verilog HDL, Xilinx Vivado* March 2026

- Designed and simulated a parameterised **Traffic Light Controller** using **Moore FSM architecture** for automated single-junction traffic management.
- Programmed RTL-based 3-state transition logic and timer-driven signal sequencing, successfully synchronising RED, GREEN, and YELLOW phases to reduce simulated traffic delay.
- Optimized traffic signal sequencing and state transitions, identifying 3 critical logic flaws that ensured accurate and stable controller operations for traffic systems.

Verilog Implementation for Digital Circuits | *Verilog HDL, Xilinx Vivado* January 2026

- Developed 20+** combinational and sequential digital circuits (MUX, decoders, flip-flops) using Verilog HDL to optimise logic gate efficiency.
- Executed** rigorous functional verification on 20+ digital circuits, reducing simulation errors by 95%.

ACHIEVEMENTS

- Qualified **GATE EC 2026** with a GATE Score of 349.
- Awarded a **Rs. 50,000 stipend** during internship at **Fact App** based on technical performance and contribution.
- Certifications:** MATLAB Onramp (MathWorks) Introduction to Internet of Things (NPTEL – IIT KGP), Data Analytics (SWAYAM – NITTTR), Introduction to Intellectual Property (SWAYAM – IIT KGP),

EXTRACURRICULAR ACTIVITIES

- Led a 3-member team as **Content Manager Lead** at **Equilibrio Techfest, GGV**.
- Progressed from **Content Team Member** to **Content Team Lead**, managing a team of 5, then promoted to **Student Coordinator**, managing newsletter workflows, team coordination, and student activities at **The Flip Flops**.
- Mentored 50+ students through **EDUGLE**, doubt solving, providing academic guidance and counselling support with peers from IITs, NITs, and IIITs.